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1 November 1973
SUPERSEDING
MIL-T-28721A(EC)
18 May 1972
(See 6.4)

# MILITARY SPECIFICATION

# TELEPHONE SET, TA-840( )/U

## 1. SCOPE

1.1 This specification covers the performance, production and testing requirements for a Telephone Set TA-840( )/U. This set is for use with shipboard secure voice communications systems.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-S-901 - Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for
MIL-E-16400 - Electronic Equipment, Naval Ship and Shore: General Specification
MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Repair Parts: Packaging and Packing of
MIL-I-45208 - Inspection System Requirements

STANDARDS

FEDERAL

FED-STD-595 - Colors

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-108 - Definition of and Basic Requirement for Enclosures for Electric and Electronic Equipment
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurements of
MIL-STD-471 - Maintainability/Verification/Demonstration Evaluation
MIL-STD-781 - Reliability Tests: Exponential Distribution
MIL-STD-785 - Reliability Program for Systems and Equipment Development and Production
MS 3102 - Connector, Receptacle, Electric, Box Mounting Solder Contacts, AN Type



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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General requirements. The equipment shall meet the requirements of MIL-E-16400 to the extent specified herein.

3.1.1 Equipment composition. The telephone set shall consist of a handset and a base.

3.1.1.1 Handset. The handset with cable shall consist of the following:

3.1.1.1.1 Transmitter. The transmitter shall consist of a microphone and a microphone amplifier.

3.1.1.1.2 Receiver. See 3.12.1.2.

3.1.1.1.3 PTT (push-to-talk) switch. See 3.12.1.4.

3.1.1.1.4 Cable. The cable shall be terminated in a connector for connection to the base.

3.1.1.2 Base. The base shall contain the necessary controls, indicators, switches and circuitry required to meet the performance requirements of this specification. A cradle shall be provided on the front of the base for secure stowage of the handset when it is not in use. The cradle shall contain a hook switch.

3.1.2 Equipment description. The telephone set when connected to external equipment will be part of a secure voice system. The secure voice system shall be capable of communicating in a CIPHER or PLAIN mode of operation. The mode of operation shall be selected at the telephone set.

3.2 First article sample. Prior to beginning production, a sample shall be tested as specified in 4.3 (see 6.3).

3.3 Finish. The equipment shall be in accordance with the requirements of the coating and treatments paragraph of MIL-E-16400. The finish shall be red enamel, No. 11136 of FED-STD-595.

3.4 Safety. The equipment shall be in accordance with the requirements of the safety paragraph of MIL-E-16400.

3.5 Workmanship. Workmanship shall be in accordance with MIL-E-16400.

3.6 Environmental conditions.

3.6.1 Temperature. Temperature conditions shall be in accordance with Class 4 of MIL-E-16400, except the lower limit in the nonoperating temperature condition shall be -55°C.



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3.6.2 Humidity. The equipment shall meet the humidity requirements of MIL-E-16400.

3.7 Selection of parts and materials. Parts and material shall be selected in accordance with MIL-E-16400.

3.8 Electrical design. The equipment shall meet the following electrical design requirements.

3.8.1 Primary power. The equipment shall be designed to operate within the requirements of this specification when the primary power supply line voltage is 24 volts ±10 percent DC. Current required shall not exceed 150 milliamperes (mA).

3.8.1.1 Transient voltage. The equipment shall meet the transient voltage requirements of MIL-E-16400.

3.8.2 Electrical bonding. The equipment shall meet the requirements of MIL-E-16400.

3.8.3 Internal cabling. The equipment shall meet the requirements of MIL-E-16400.

3.8.4 Wire terminations. The equipment shall meet the requirements of MIL-E-16400.

3.8.5 Identification of conductors. The equipment shall meet the requirements of MIL-E-16400.

3.9 Test points. Test points shall be in accordance with MIL-E-16400.

3.10 Mechanical design. The base shall use plug-in printed circuit cards.

3.10.1 Construction.

3.10.1.1 Base. The base shall be constructed of ferrous metal and shall have a hinged front panel. The degree of enclosure shall be dripproof (15 degrees) in accordance with MIL-STD-108.

3.10.1.2 Handset cable. The handset cable shall be shielded. The shielding coverage shall be a minimum of 90 percent. The ends shall be securely fastened and stress relieved to preclude failure with normal usage.

3.10.1.3 Physical requirements. The base shall be designed for bulkhead mounting. The dimensions shall not exceed 10 inches (254 mm) high, 6 inches (152.4 mm) wide and 3 inches (76.2 mm) deep excluding handset, controls, hardware and cover plates mounted on the base. Overall depth with handset mounted in the cradle shall not exceed 6.5 inches (165.1 mm). Total weight shall not exceed 7 pounds (3.175 kg).

3.10.1.4 Cable entrances. Cable entrances shall be provided on top, bottom and the right side of the base. The cradle shall be mounted on the left side of the front panel.



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3.10.2 Welding and soldering. Welding and soldering shall be in accordance with the requirements of MIL-E-16400.

3.10.3 Shock. Shock requirements shall be in accordance with Grade A, Class 1, Type A lightweight equipment as specified in MIL-S-901.

3.10.3.1 Shock test procedures. The shock test procedures shall be as specified in MIL-E-16400. The handset may be separately secured to the test plate to preclude damage resulting from its detachment from the telephone handset hanger (cradle).

3.10.4 Vibration. The equipment shall meet the vibration requirements of MIL-E-16400.

3.10.5 Hinged cover. The equipment shall meet the requirements of MIL-E-16400.

3.10.6 Fastening devices. The equipment shall meet the requirements of MIL-E-16400.

3.11 Controls, indicators and panel layout.

3.11.1 Controls and indicators. The following controls and indicators shall be provided on the front panel.

3.11.1.1 "Speaker Muted By" switch. See 3.12.2.6.

3.11.1.2 Lamp dimmer control. The lamp dimmer control shall permit the operator to adjust the intensity of the detect, disconnect, cipher and plain lamps simultaneously.

3.11.1.3 Cipher switch. See 3.12.2.5.1.

3.11.1.4 Plain switch. See 3.12.2.5.1.

3.11.1.5 Detect lamp. The detect lamp shall provide the operator a visual indication of the receive cipher condition.

3.11.1.6 Disconnect lamp. The disconnect lamp shall provide the operator a visual indication when the telephone set does not have access to the external circuit.

3.11.1.7 Cipher lamp. See 3.12.2.5.2.

3.11.1.8 Plain lamp. See 3.12.2.5.2.

3.11.2 Panel layout. The base front panel layout shall be identical to that of the Telephone Set TA-840/U.



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3.12 Functional requirements.

3.12.1 Handset.

3.12.1.1 Transmitter.

3.12.1.1.1 Microphone. The microphone shall be a dynamic type with a noise canceling feature and an impedance of 15 ohms plus or minus 20 percent. The output of the microphone shall be applied to the microphone amplifier which shall be shielded.

3.12.1.1.2 Electrical characteristics. The combined microphone and microphone amplifier shall conform to the following characteristics:

(a) Output level - 20 dBm average, 0 dBm peak.
(b) Frequency response - Within 8 dB peak-to-peak from 70 to 4000 Hz.
(c) Output impedance - 600 ohms plus or minus 10 percent at 1 kHz unbalanced.
(d) Distortion - At a sustained peak output level of 0 dBm the microphone and microphone amplifier shall not introduce audio distortion in the frequency range 70 to 4000 Hz greater than 5 percent.

3.12.1.2 Receiver. The receiver shall conform to the following characteristics:

(a) Maximum level - Accept a 12 dBm maximum power input without permanent damage.
(b) Operating level - 100 millivolts rms nominal at the input.
(c) Distortion - With a voltage of 100 millivolts rms at the receiver input terminals, the receiver shall not introduce more than 5 percent distortion in the acoustic wave at any frequency between 200 and 4000 Hz.
(d) Frequency response - Within 12 dB peak-to-peak from 200 to 4000 Hz.
(e) Impedance - 15 ohms ±20 percent at 1 kHz and 100 millivolts rms at the input terminals, unbalanced to ground.

3.12.1.3 Cable connector. The handset shall be connected to the base through a connector in accordance with MS3102, Class A, Size 16, Arrangement 1, Style S.

3.12.1.3.1 The connector shall conform to the following pin connections:

(a) Pins A and B - Microphone amplifier output. Pin B shall be at signal ground when connected to the base.
(b) Pins C and D - Receiver input. Pin D shall be at signal ground when connected to the base.
(c) Pins E and F - PTT switch.
(d) Pin F - Chassis ground.



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3.12.1.4 PTT switch. The PTT switch shall be a normally open push button switch and shall conform to the following characteristics:

(a) Resistance - 100 ohms MAX switch closed
100 k ohms MIN switch open
(b) Voltage drop (switch closed) - 1 volt DC MAX
(c) Open circuit switch voltage - 30 volts DC MAX
(d) Closed circuit current - 120 milliamperes DC MAX

3.12.2 Base.

3.12.2.1 Terminal boards. The base shall interface to equipments external to telephone set through two terminal boards, TB1 and TB2. These interfaces shall conform to the following requirements:

(a) TB1 -

| Terminal No. | Function |
|---|---|
| 2 and 3 | Transmitter audio output (see 3.12.2.2) |
| 4 and 5 | Receive audio input (see 3.12.2.3) |
| 6 | PTT (see 3.12.2.4) |
| 7 | Mode select (see 3.12.2.5) |

(b) TB2 -

| Terminal No. | Function |
|---|---|
| 1 and 2 | Audio to speaker amp (see 3.12.2.6) |
| 3 | +24vdc |
| 4 | Hook switch (see 3.12.2.7) |
| 5 | Signal ground (see 3.12.2.8) |
| 6 | Mode indicate (see 3.12.2.5) |

3.12.2.2 Transmit audio output. Capability for transmitting audio output shall be enabled when the push-to-talk switch is actuated. The transmit audio output shall conform to the following:

(a) Level - -20 dBm average, 0 dBm peak.
(b) Frequency response - 8 dB, peak-to-peak from 70 to 4000 Hz.
(c) Impedance - 600 ohms ±10 percent balanced.

3.12.2.3 Receive audio input. The receive audio input shall conform to the following characteristics:

(a) Input level - 0 dBm average, +10 dBm MAXIMUM
(b) Frequency response - 12 dB peak-to-peak from 200 to 4000 Hz
(c) Impedance - 600 ohms ±10 percent balanced.

3.12.2.4 PTT function. A relay-operated pair of contacts shall be provided which place terminal No. 6 of TB1 at signal ground when the PTT switch on the handset is closed and the handset is connected to the base. When the PTT switch is open terminal No. 6 of TB1 shall be open. The relay operated contacts shall conform to the following characteristics:

(a) Resistance - 100 ohms MAX (contacts closed)
100 ohms MIN (contacts open)
(b) Voltage drop (contacts closed) - 1 volt MAX DC



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3.12.2.5 Modes of operation.

3.12.2.5.1 Mode selection. Selection of CIPHER or PLAIN mode of operation shall be made by means of normally open spring-loaded push-button switches. Depressing the CIPHER switch shall place a ground at terminal No. 7 of TB1. Depressing the PLAIN switch shall place +24 volts DC to terminal No. 7 of TB1. The CIPHER and PLAIN switches shall be of the illuminated type. The interface at terminal No. 7 of TB1 shall conform to the following characteristics:

(a) Resistance - 100 ohms MAX (CIPHER SW closed)
200 ohms MAX (PLAIN SW closed)
100 k ohms MIN (CIPHER and PLAIN switches open)
(b) Voltage drop - 1 volt DC MAX (PLAIN or CIPHER SW closed)

3.12.2.5.2 Mode indication. Indication of the mode of operation shall be as follows:

(a) CIPHER MODE - When terminal No. 6 of TB2 is placed at signal ground external to the base, the CIPHER push-button switch shall illuminate. Color shall be green. Resistance of the external circuit is 200 ohms MAX.
(b) PLAIN MODE - When 24 VDC from an external circuit is applied to terminal No. 6 of TB2 the PLAIN push-button switch shall illuminate. Color shall be red. Resistance of the external circuit is 200 ohms MAX.

3.12.2.6 Auxiliary audio output. The auxiliary audio output circuit shall provide a bridging output. A three position rotary switch labeled "Speaker Muted By" shall be provided for controlling the output of the auxiliary circuit. The auxiliary circuit shall conform to the following characteristics:

(a) Output (from receive audio input) is enabled by the following conditions:

(1) "Speaker Muted By" switch in HK SW position and handset off hook.
(2) "Speaker Muted By" switch in PTT position and push-to-talk is not activated.

(b) Output is disabled when "Speaker Muted By" switch is off.
(c) Output Level - +10 dBm MAX
(d) Impedance - Bridging (greater than 100 k ohms) looking toward external circuit. NOTE: The telephone set shall function with no greater degradation than 6 dB if external impedance connected to the auxiliary output is 600 ohms ±20 percent.

3.12.2.7 Hook switch. The hook switch shall be a double pole single throw type and shall function as follows:

(a) Provide a switch closure to signal ground at terminal No. 4 of TB2 when handset is taken off hook and shall conform to the following characteristics:



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(1) Open circuit (handset ON hook) - resistance at terminal No. 4 - 100 k ohms MIN
(2) Closed circuit (handset OFF hook) - resistance at terminal No. 4 - 100 ohms MAX

(b) Provide a switch closure which connects +24 VDC to pin E of the base connector when the handset is taken off hook.

3.12.2.8 Signal ground. Signal ground shall interface to TB2 terminal No. 5. Chassis ground shall not be connected to signal ground. Provision shall be made to internally strap signal ground to chassis ground when desired. When unstrapped the resistance from signal ground to chassis ground shall be one megohm minimum and the capacitance shall be 2000 picofarad maximum.

3.12.2.9 Phantom circuits. The following phantom circuits shall be supplied.

3.12.2.9.1 Detect. A phantom circuit with respect to signal ground shall be provided to indicate the receive cipher condition. On the application of 6 volts DC ±20 percent through 6.8 k ohms ±20 percent to the phantom (via terminals No. 2 and No. 3 of TB1) the DETECT lamp on the front panel shall be illuminated. The color shall be amber. On application of 0 volts DC ±0.5 volts the DETECT lamp shall not be illuminated.

3.12.2.9.2 Disconnect. A phantom circuit with respect to signal ground shall be provided to indicate connect-disconnect. On the application of a ground of 100 ohms or less to the phantom (via terminals No. 4 and No. 5 of TB1) the DISCONNECT lamp on the front panel shall not illuminate. On the application of an open circuit greater than 100 k ohms the DISCONNECT lamp shall illuminate. The color shall be blue.

3.12.2.10 14 VDC. When the handset is "off hook" and the PTT switch is depressed 14 volts DC ±10 percent shall be supplied through a series resistance of 470 ohms ±20 percent to terminals A and B of the receptacle which connects to the handset connector.

3.13 Reliability.

3.13.1 Quantitative reliability requirement. The specified MTBF ($\theta_o$ is defined by MIL-STD-781) of this equipment shall be 10,000 hours at 50°C ambient.

3.13.2 Reliability program. The contractor shall prepare an effective reliability program in accordance with MIL-STD-785 (see 6.2.2). The program shall ensure reliability involvement throughout all aspects of the design, development and production as necessary to meet the contractual reliability requirements.

3.14 Maintainability.

3.13.1 Quantitative maintainability requirement. The equipment shall have a specified equipment repair time (ERT) of 30 minutes with repair accomplished



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by the replacement of modules, chassis-mounted electronics, and electromechanical parts.

3.15 Electromagnetic compatibility (EMC). The handset shall meet the Electromagnetic interference (EMI)/EMC requirements of Class IC equipment in accordance with MIL-STD-461.

3.15.1 Limit requirements. The following modifications to the limit requirements of MIL-STD-461 shall apply:

(a) Test CS02 frequency range shall be extended to 10 Gigahertz (GHz).
(b) Test CS03 shall only be performed up to 50 kHz.
(c) Tests CS04, CS05, CS08, RE04 and RS04 shall not apply.
(d) Test RS03 shall be performed at 50 volts per meter in the frequency range 200-400 megahertz (MHz). The test limit of MIL-STD-461 shall apply for the remaining frequencies specified.
(e) For susceptibility test purposes a 500 microvolt signal induced in the microphone or a -10 dBm signal in the earpiece shall be construed as a failure.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection system requirement. The contractor shall provide and maintain a written inspection system acceptable to the Government for supplies and services covered by this specification. The inspection system shall be in accordance with MIL-I-45208.

4.1.2 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of a) surveillance of the operations to determine that practices, methods and procedures of the written system description are being properly applied, b) Government product inspection to measure quality of product to be offered for acceptance, and c) Government product inspection of delivered products to assure compliance with all requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated.

4.1.3 Quality assurance terms and definitions. Quality assurance terms used in this specification shall be as defined in MIL-STD-109.



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4.2 Classification of inspection. The method of examination and testing of the equipment fall within the following classifications:

(a) First article inspection (see 4.3)
(b) Quality conformance inspection

(1) Production inspection (see 4.4.1)
(2) Production control inspection (see 4.4.2)

(c) Reliability tests (see 4.5)
(d) Maintainability demonstration (see 4.6)
(e) Inspection of preparation for delivery (see 4.7)

4.3 First article inspection. Unless otherwise specified (see 6.2.1), one unit shall be required for first article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the tests specified in table 1.

Table I. First article inspection

| Inspection | Requirement Paragraph | Test Paragraph |
|---|---|---|
| Surface examination | 3.1, 3.3, 3.5, 3.7, 3.8.2, 3.8.3, 3.8.4, 3.8.5, 3.10, 3.10.1.1, 3.10.1.2, 3.10.1.3, 3.10.1.4, 3.10.2, 3.11.1, 3.11.2, 3.12.1.3, 3.12.2.8 | 4.8.1 |
| Operating test | 3.4, 3.12 | 4.8.2 |
| Weights and dimensions | 3.10.1.3 | 4.8.1 |
| Supply line voltage | 3.8.1, 3.8.1.1 | 4.8.1 |
| Heat test | 3.6.1 | 4.8.1 |
| Enclosure test 1/ | 3.10.1.1 | 4.8.1 |
| Temperature | 3.6.1 | 4.8.1 |
| Humidity | 3.6.2 | 4.8.1 |
| Power | 3.8.1 | 4.8.1 |
| Controls & control circuit | 3.12.1.4, 3.12.2.4, 3.12.2.5, 3.12.2.6a, 3.12.2.6b, 3.12.2.9 | 4.8.1 |
| Shock | 3.10.3 | 4.8.1 |

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Table I. First article inspection (continued)

| Inspection | Requirement Paragraph | Test Paragraph |
|---|---|---|
| Vibration | 3.10.4 | 4.8.1 |
| EMI/EMC susceptibility test | 3.15 | 4.8.3 |
| Salt spray | 3.3 | 4.8.1 |

1/ Enclosure test shall be conducted after completion of shock/ vibration.

4.4 Quality conformance inspection.

4.4.1 Production inspection. Production inspection shall be made on every equipment offered for delivery. The inspection shall comprise such examination and testing as will prove the workmanship and reveal the omissions and errors of the production process such as functional and performance tests at a limited number of points, tests which detect deviations from design, tests of adjustment, and tests which detect hidden defects of material. Production inspection shall consist of the tests approved in accordance with applicable test procedures and shall include the inspections of table II.

Table II. Production inspection

| Inspection | Requirement Paragraph | Test Paragraph |
|---|---|---|
| Surface examination | 3.3, 3.5, 3.10.1.4, 3.12.1.3, 3.12.1.3.1d | 4.8.1 |
| Operating test | 3.4, 3.12 | 4.8.2 |

4.4.2 Production control inspection. Production control inspection, including sampling, shall conform to table III and to the procedures for special inspection levels of MIL-STD-105. The inspection level shall be S-4 for normal, tightened, and reduced inspection. Production control inspection shall be performed on equipments that have passed production inspection. The equipment shall satisfactorily meet the requirements of production control inspection prior to release for shipment.

Table III. Production control inspection

| Inspection | Requirement paragraph | Test paragraph | AQL percent |
|---|---|---|---|
| Weights & dimensions | 3.10.1.3 | 4.8.1 | 10 |
| Supply line voltage | 3.8.1, 3.8.1.1 | 4.8.1 | 6.5 |



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Table III. Production control inspection (continued)

| Inspection | Requirement paragraph | Test paragraph | AQL percent |
|---|---|---|---|
| Heat test | 3.6.1 | 4.8.1 | 10 |
| Enclosure | 3.10.1.1 | 4.8.1 | 10 |
| Power | 3.8.1 | 4.8.1 | 6.5 |
| Control & control circuits | 3.12.1.4, 3.12.2.9 3.12.2.4, 3.12.2.5, 3.12.2.6(a) and (b) | 4.8.1 | 6.5 |

4.4.2.1 Rejected lots. If an inspection lot is rejected, the supplier may withdraw the lot from further inspection. The supplier may also rework a rejected lot to correct the defective units and reinspect the lot using tightened inspection. Rejected lots shall be kept separate from new lots and shall not lose their identity.

4.4.2.2 Reinspection of conforming production control test units. Unless otherwise specified, sample units which have been subject to and passed production control inspections may be accepted on the contract provided they are refurbished to like new condition, resubjected to and pass production inspection.

4.5 Reliability tests.

4.5.1 Reliability demonstration. Reliability demonstration shall be performed in accordance with MIL-STD-781.

4.5.2 Reliability qualification tests. Qualification testing shall be performed on at least twenty production units at test level C with test plan V, proceeding to a decision in accordance with the criteria specified in MIL-STD-781 for test plan V.

4.5.3 Reliability sampling tests. Reliability sampling tests shall be conducted throughout production to ensure continued production of acceptable equipment. Sampling testing shall be at test level C, with test plan VI of MIL-STD-781. At least three units shall be randomly selected for each month's production for reliability sampling tests. Testing shall proceed to a decision in accordance with the criteria specified in MIL-STD-781 for test plan VI.

4.5.4 Failure. Failure shall be defined as any malfunction or parameter deviation that prevents the equipment from performing within the operational requirements set forth in this specification.

4.5.5 Reliability test procedures. The reliability tests shall be conducted using the procedures, schedules and performance limits specified herein as a minimum.



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4.5.6 Burn-in. The contractor may elect to operate the reliability test samples for a time prior to the initiation of reliability testing. Any such burn-in applied to test samples must be applied to all items of equipment submitted for acceptance. Failures occurring during this period may be considered nonrelevant but shall be recorded and reported. Burn-in time for all items of equipment submitted for acceptance shall be at the same levels of environment and for the same integral number of cycles as the burn-in conducted on the test samples.

4.6 Maintainability demonstration. The equipment maintainability requirement shall be demonstrated in accordance with Method 3 of MIL-STD-471. The maintainability demonstration shall be successfully completed prior to the delivery of any equipment. The maintainability demonstration shall be performed after completion of first article testing and prior to reliability testing.

4.7 Inspection of preparation for delivery. Inspection shall be conducted to ensure conformance with the requirements of section 5.

4.8 Inspection procedures.

4.8.1 Tests and examinations. Tests and examinations specified herein shall be conducted in accordance with the applicable inspection procedures of MIL-E-16400, except as modified below.

4.8.2 Operating test. The equipment shall be energized and subjected to an operating test to insure the proper functioning and performance at an adequate number of points in the required range to demonstrate conformance to specification requirements, including safety. The operating test shall include, but not be limited to, the tests shown in table IV.

Table IV. Operating test

| Test | Requirement reference |
|---|---|
| Handset microphone impedance and frequency response | 3.12.1.1.1, 3.12.1.1.2(b) |
| Handset receiver impedance and frequency response | 3.12.1.2(d) and (e) |
| Telephone/Handset interface | |
| Input/Output Characteristics | 3.12.1.1.2, 3.12.1.2 |
| Push-to-Talk | 3.12.1.4 |
| 14 VDC | 3.12.2.10 |
| Terminal Board interface | |
| Audio input/output Characteristic | 3.12.2.2, 3.12.2.3 |
| Push-to-Talk | 3.12.2.4 |



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Table IV. Operating test (continued)

| Test | Requirement reference |
|---|---|
| Mode select/identification | 3.12.2.5 |
| Auxiliary audio output | 3.12.2.6 |
| Hook Switch | 3.12.2.7 |
| Signal Ground | 3.12.2.8 |
| Phantom circuits | 3.12.2.9 |

4.8.3 EMI/EMC susceptibility test. The EMI/EMC tests shall be performed in accordance with MIL-STD-462 modified as follows:

(a) The test CS02 frequency range shall be extended to 10 GHz and the signal shall be pulse modulated with a 2 µs pulse at 400 or 100 PPS, or 50 percent AM at 1 kHz.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation and packaging. Preservation and packaging shall be as specified (see 6.2.1) in MIL-E-17555 or applicable specification.

6. NOTES

6.1 Intended use. This equipment is intended for use in a secure voice communications system on all ship types. It will provide an operator the capability to transmit either in a secure mode (CIPHER) or in a PLAIN mode when connected to the radio channel. Reception will be in either the CIPHER or PLAIN mode, automatically selected by associated external equipment.

6.2 Ordering data. Procurement documents should specify the following:

6.2.1 Procurement requirements.

(a) Title, number and date of this specification
(b) Number of first article samples to be submitted if other than specified in 4.3
(c) Levels of preservation-packaging and packing, and applicable marking (see 5.1)



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6.2.2 Contract data requirements. Data generated by this document are not deliverable unless specified on the Contract Data Requirements List (DD Form 1423) or the contract schedule. The data required by this specification include, but are not restricted to the following:

(a) Reliability program (see 3.13.2)

6.3 First article. Invitations for bids should provide that the Government reserves the right to waive the requirement for first article samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.4 CHANGES FROM PREVIOUS ISSUE. THE EXTENT OF CHANGES (DELETIONS, ADDITIONS, ETC.) PRECLUDE THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT.

Preparing activity:
Navy - EC

(Project 5805-N216)



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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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